

Plant Disease Recognition Model Analysis

A Comparative Study: EfficientNet-B0 vs. ResNet50

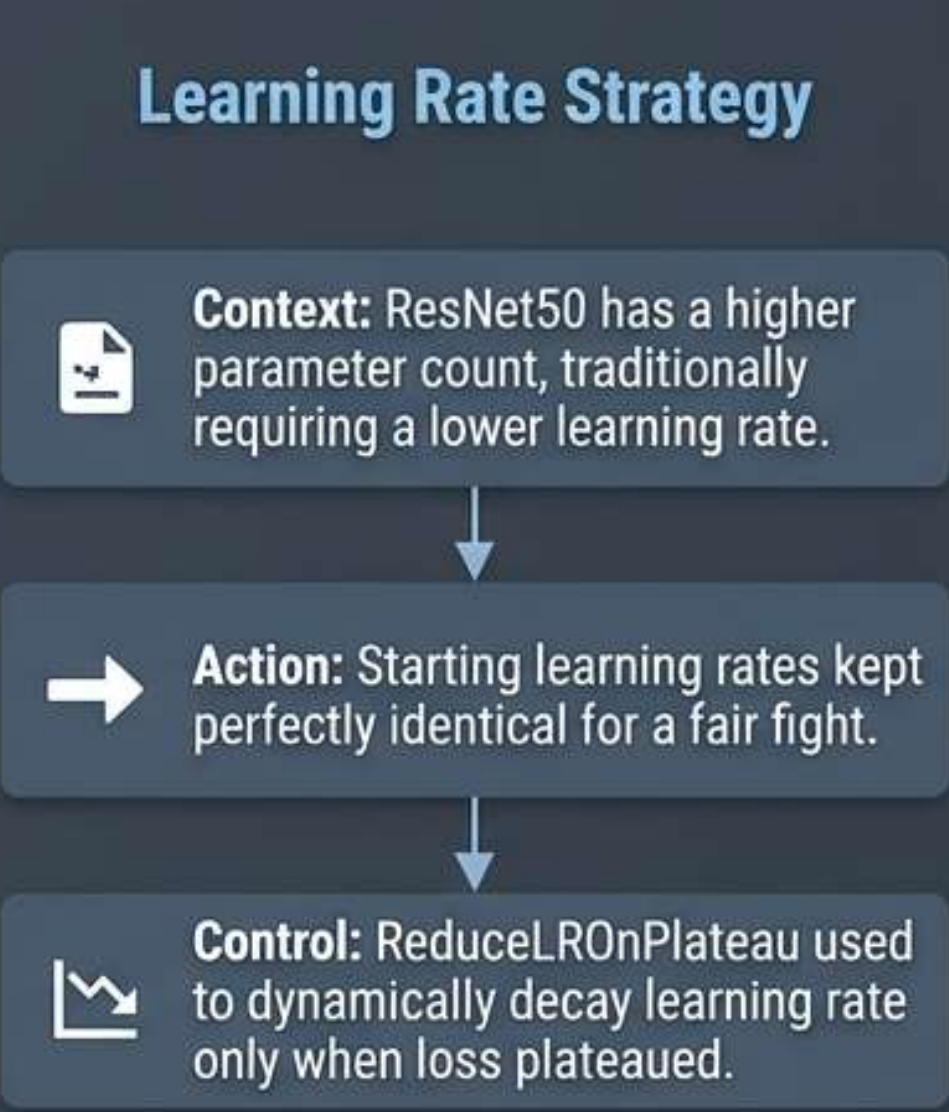
Dataset Distribution

Class	Total	Train	Validation	Test
 Healthy	458	320	69	69
 Powdery Mildew	430	301	64	65
 Rust	434	304	65	65
Total	1322	925	198	199

Transformations Pipeline



The Fairness Protocol



EfficientNet-B0

95.33%

Accuracy

84.22%

Avg Confidence

Precision: **0.95**

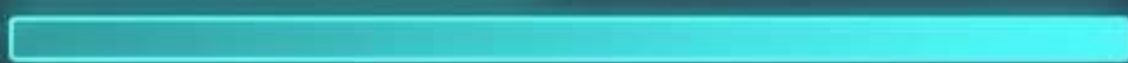
Recall: **0.95**

F1-Score: **0.95**

👍 TP: 143



👎 TN: 293



⊗ FP: 7



🚫 FN: 7



ResNet50

97.33%

Accuracy

84.42%

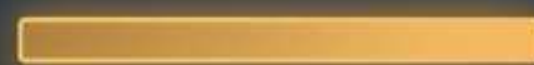
Avg Confidence

Precision: **0.97**

Recall: **0.97**

F1-Score: **0.97**

👍 TP: 146



👎 TN: 296



⊗ FP: 4



🚫 FN: 4



Diagnostic Key

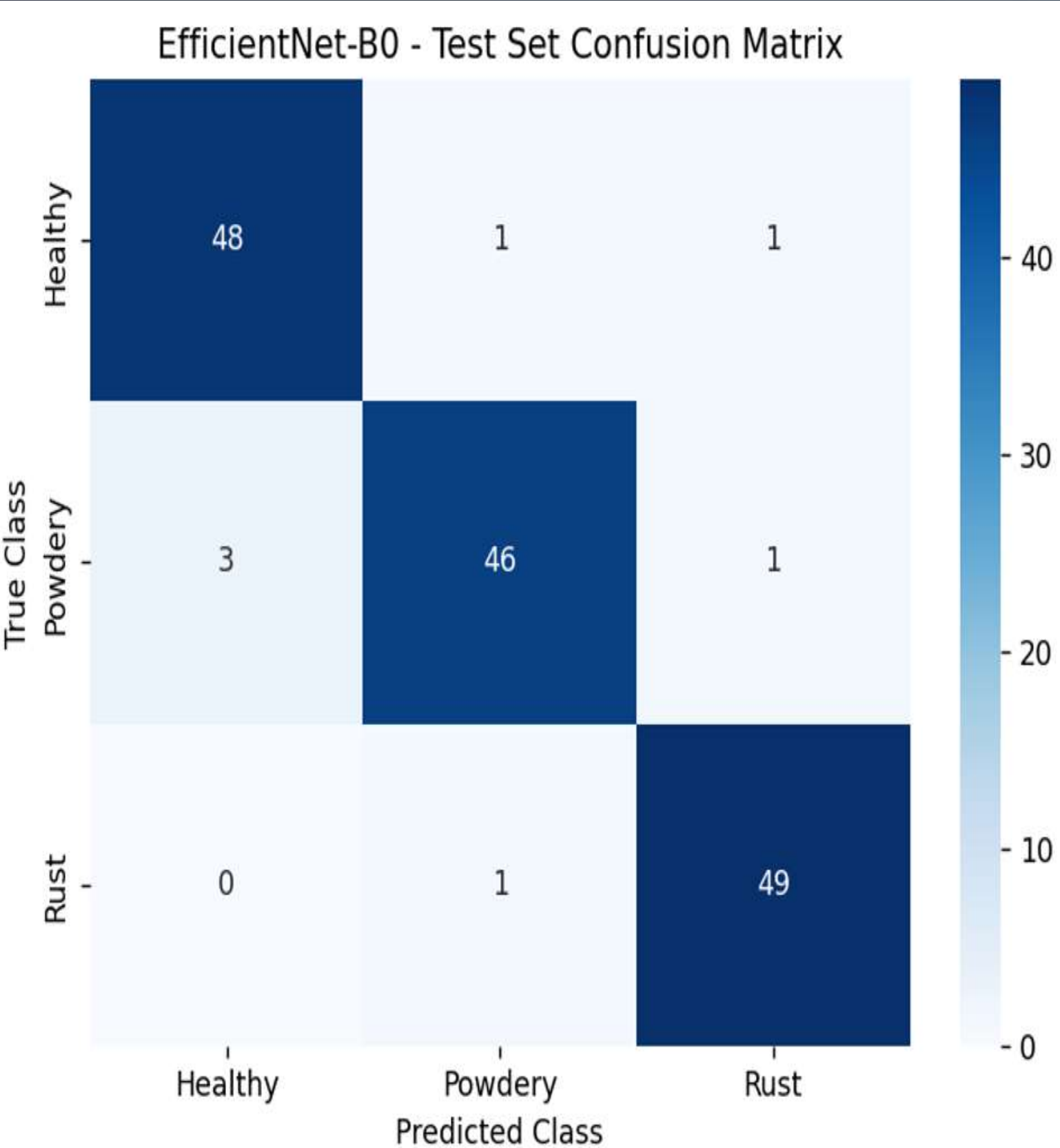
TP (True Positive): Diseased plant correctly identified as diseased.

FP (False Positive): Healthy plant incorrectly identified as diseased.

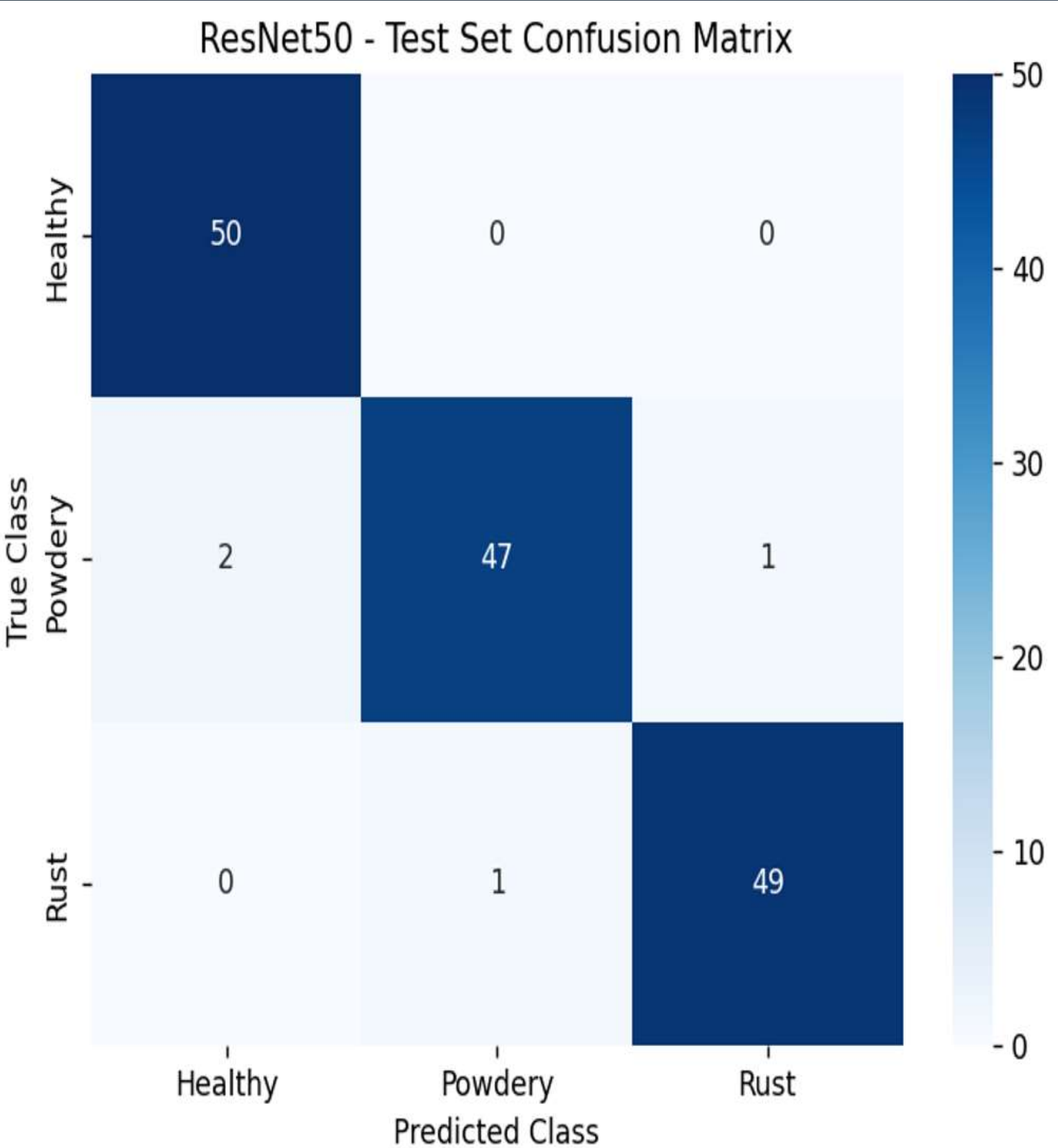
TN (True Negative): Healthy plant - correctly identified as healthy.

FN (False Negative): Diseased plant incorrectly identified as healthy.

EfficientNet-B0 Confusion Matrix

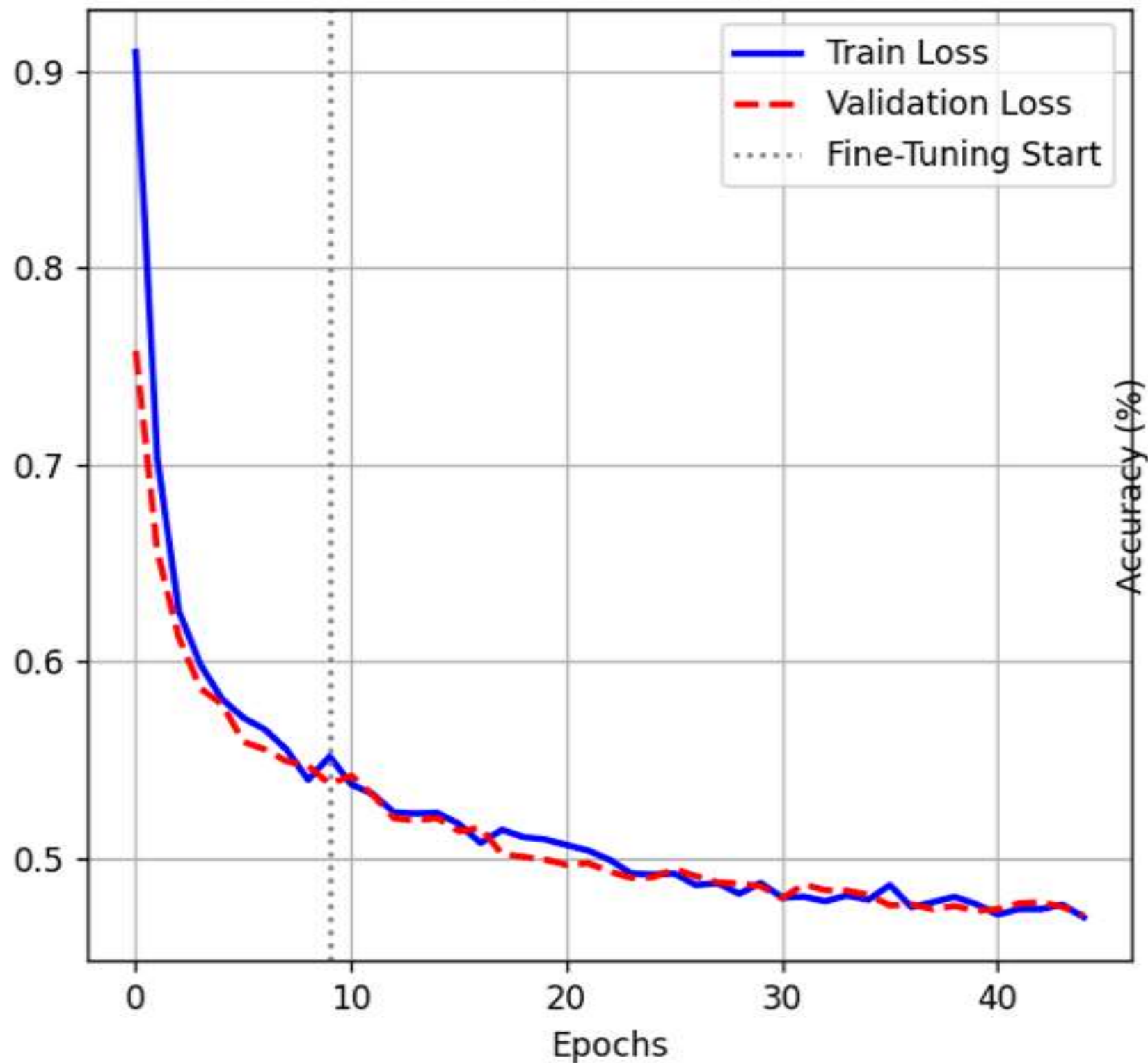


ResNet50 Confusion Matrix

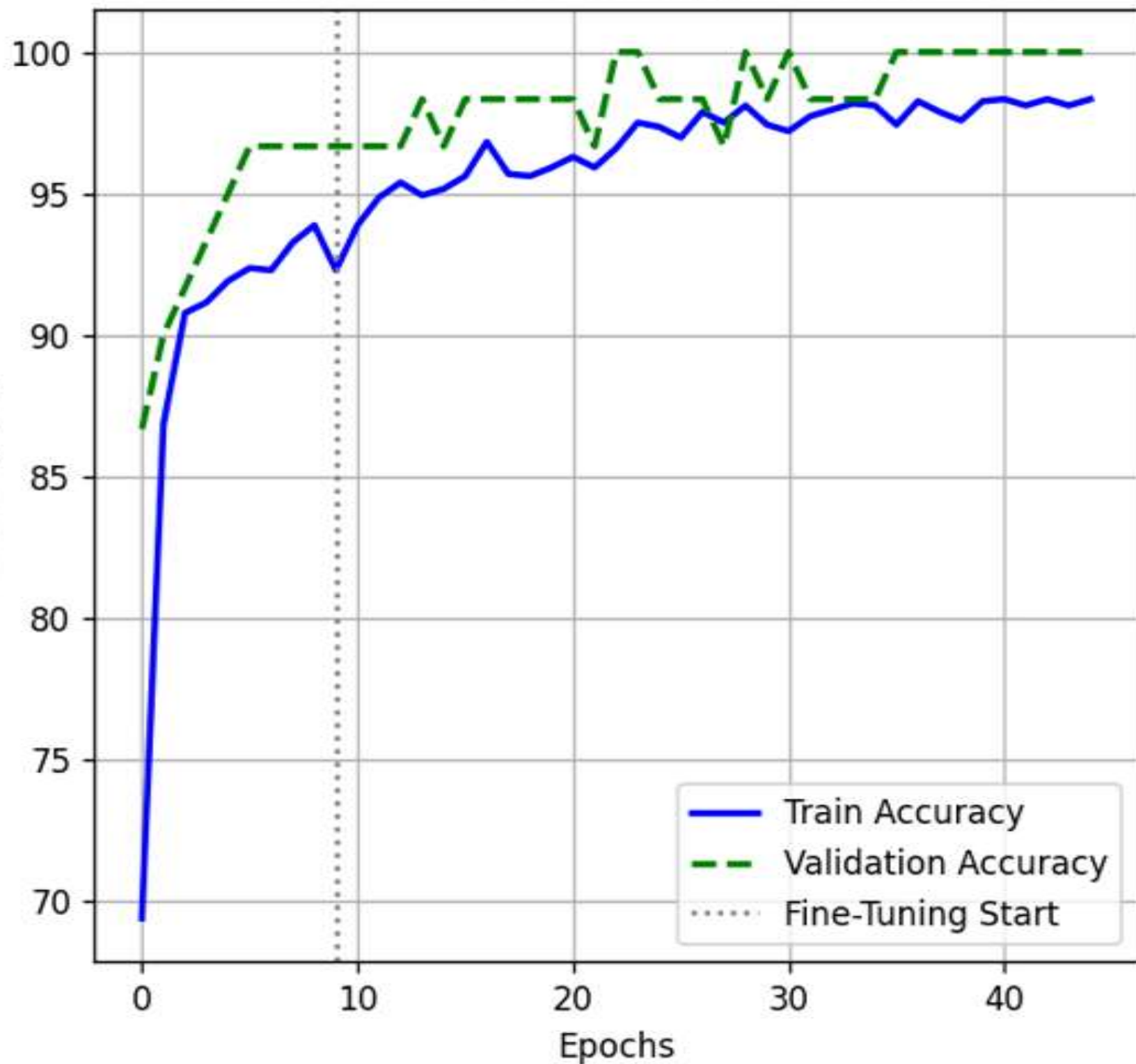


EfficientNet-B0 Dynamics

EfficientNet-B0 (Fine-Tuning): Loss Curve

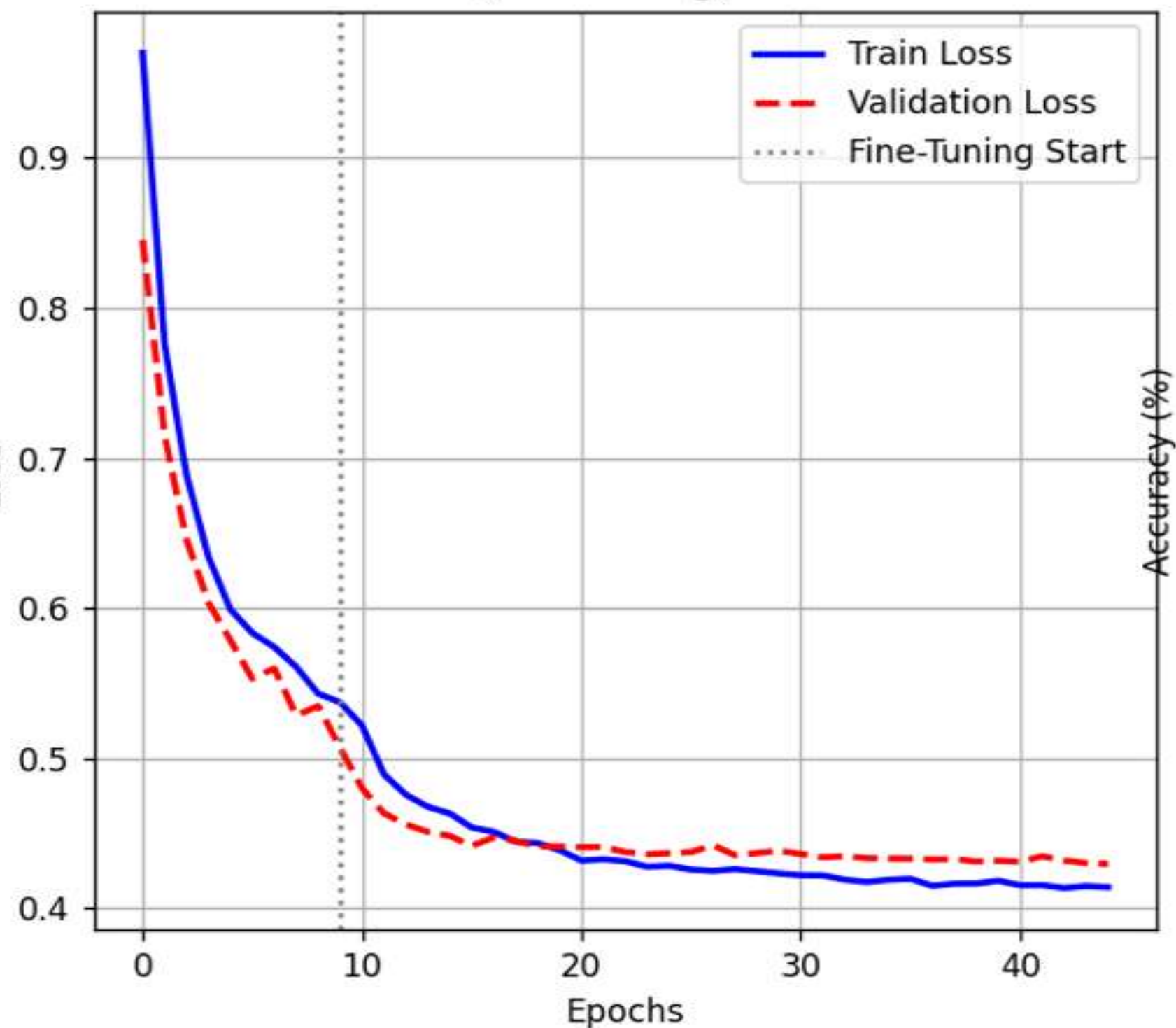


EfficientNet-B0 (Fine-Tuning): Accuracy Curve

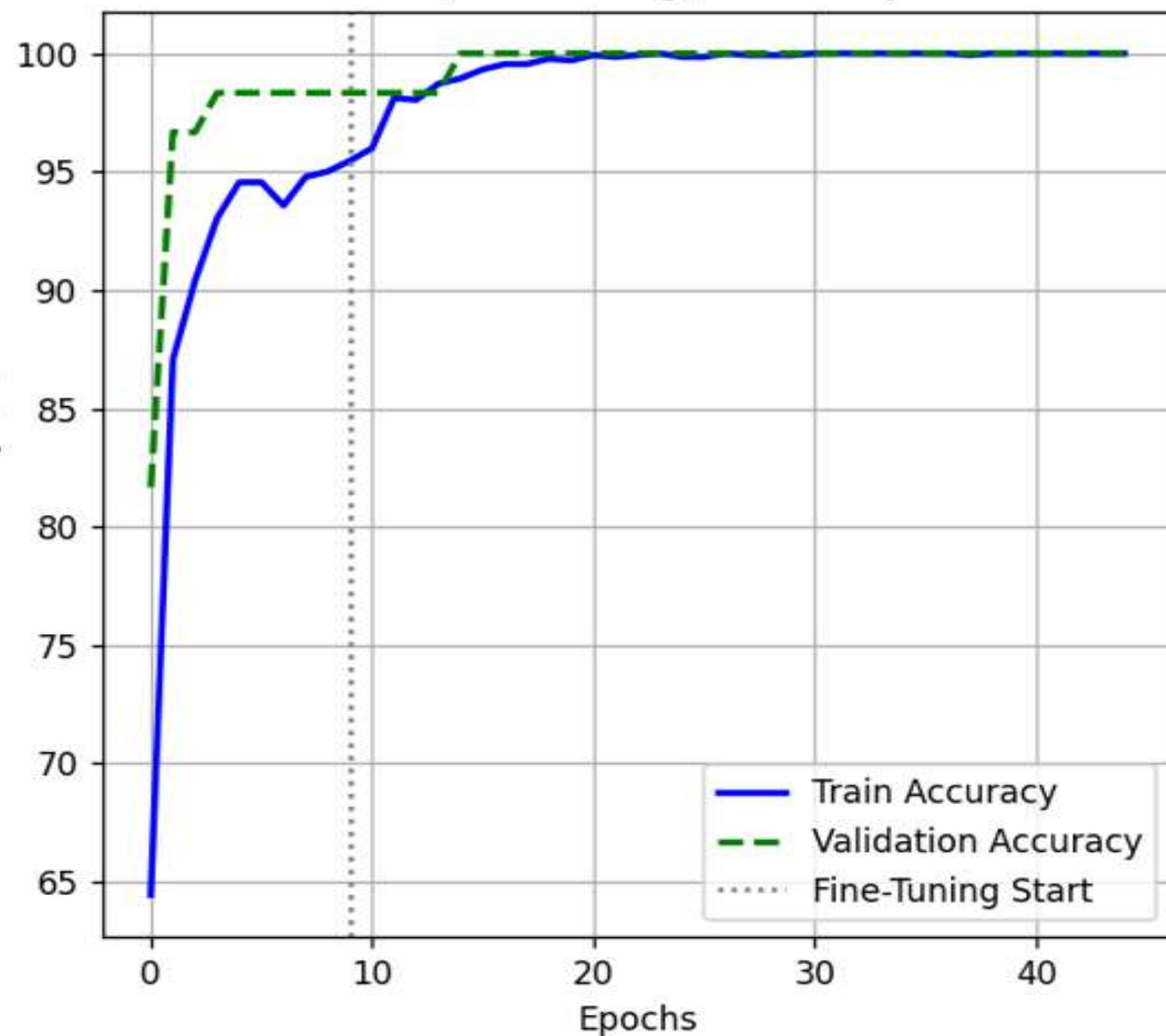


ResNet50 Dynamics

ResNet50 (Fine-Tuning): Loss Curve



ResNet50 (Fine-Tuning): Accuracy Curve



Architecture Comparison & Final Verdict

Metric	EfficientNet-B0	ResNet50	Winner
Test Accuracy	95.33%	🏆 97.33%	🏆 ResNet50
Error Count	7/150	🏆 4/150	🏆 ResNet50
Average Confidence	84.22%	🏆 84.42%	🏆 ResNet50
Speed per Epoch	⚡ ~2.0 sec 2x Faster	~4.1 sec	⚡ EfficientNet-B0
Overfitting Resilience	Excellent	Excellent	🤝 Tie

Verdict: ResNet50 is the overall winner with 97.33% accuracy. EfficientNet-B0 is 2x faster, but ResNet50 gives better and more reliable results.